

Ali M. Youssef Ahmed El-Helaly

Teaching Assistant | Academic Researcher in Mathematics

Department of Mathematics, Faculty of Science, Sohag University, Sohag, Egypt

alimohammedyouseuf@yahoo.com | ali2013269@science.sohag.edu.eg | g202315690@kfupm.edu.sa

+20 1117621575 | Sohag, Egypt | [LinkedIn](#) | [Homepage](#) | [YouTube Channel](#)

[ResearchGate](#) | [ORCID: 0000-0002-3219-5688](#) | [Google Scholar](#) | [Scopus](#)

Professional Summary

Teaching Assistant and mathematics researcher specializing in harmonic analysis, α -harmonic and hyperbolic harmonic functions, fractional calculus, special functions, and PDE-related analysis. Active in peer-reviewed research and international academic events, with sustained undergraduate teaching experience across core mathematical subjects.

Research Interests

Harmonic analysis; α -harmonic and hyperbolic harmonic functions; fractional calculus and fractional differential equations; special functions; complex analysis; PDE-related analysis.

Education

2023–Present **Ph.D. in Mathematics**, King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia.

Research area: Harmonic analysis, α -harmonic functions, and PDEs on the unit ball.

2020–2021 **M.Sc. in Mathematics**, Sohag University, Egypt.

Thesis: *Advanced Studies on some Higher Transcendental Functions with some Applications*.

Supervisors: Prof. Mahmoud A. Abul-Ez; Prof. Zenhom G. Kishka; Dr. Mohammed A. Seleem.

2018–2019 **Pre-Master of Science in Mathematics**, Sohag University, Egypt (Letter Grade: A^+).

2012–2016 **B.Sc. in Science (Mathematics)**, Sohag University, Egypt (Excellent with Honors; Letter Grade: A^+).

Academic Appointments

2022–Present **Teaching Assistant**, Department of Mathematics, Faculty of Science, Sohag University, Sohag, Egypt.

2016–2022 **Instructor**, Department of Mathematics, Faculty of Science, Sohag University, Sohag, Egypt.

Publications

Peer-Reviewed Journal Articles

1. **Ali Ahmed** and Adel Khalfallah.

Riesz-Fejér inequalities for hyperbolic harmonic functions in the unit ball

Riesz-Fejér inequalities for hyperbolic harmonic functions in the unit ball. Journal of Mathematical Analysis and Applications, 563:130826, 2026. doi: 10.1016/j.jmaa.2026.130826.

2. Mahmoud Abul-Ez, Mohra Zayed, and **Ali Youssef**. Further Developments of Bessel Functions via Conformable Calculus with Applications. *Journal of Function Spaces*, 2021:Article ID 6069201, 17 pages, 2021. doi: 10.1155/2021/6069201.

3. Mahmoud Abul-Ez, Mohra Zayed, and **Ali Youssef**. Further study on the conformable fractional Gauss hypergeometric function. *AIMS Mathematics*, 6(9):10130–10163, 2021. doi: 10.3934/math.2021588.

4. Mahmoud Abul-Ez, Mohra Zayed, **Ali Youssef**, and M. De la Sen. On conformable fractional Legendre polynomials and their convergence properties with applications. *Alexandria Engineering Journal*, 59(6):5231–5245, 2020. doi: 10.1016/j.aej.2020.09.052.

Conference Papers and Presentations

- (a) **Ali Youssef**, Zenhom Kishka, and Mohammed Saleem. A New Generalized Bessel Functions and Solutions of a New Class of Fractional Kinetic Equations. *Presented at the 7th Scientific Conference (2nd International) for Young Researchers, Sohag University, March 2–3, 2022.*

Selected Manuscripts / Papers under Review

- (a) **Ali Ahmed** and Adel Khalfallah. Schwarz-Type Estimates and Boundary Behavior of α -Harmonic Functions. *Manuscript under review.*
- (b) **Ali Ahmed** and Adel Khalfallah. Invariant Mean-Value Property and α -Harmonicity in the Unit Ball of $\mathbb{R}^n$. *Manuscript under review.*
- (c) **Ali Ahmed** and Adel Khalfallah. Integral Means and Sharp Riesz-Fejér Inequalities for α -Harmonic Functions in the Unit Ball. *Manuscript under review.*

Bibliometric Profile (Scopus)

<i>Scopus ID</i>	<i>Documents</i>	<i>Citations</i>	<i>h-index</i>
57219533618	4	21	3

Teaching Experience (Undergraduate Courses)

Calculus I and II; Ordinary Differential Equations; Partial Differential Equations; Probability Theory; Special Functions; Algebra; Linear Algebra; Complex Analysis; Real Analysis; Functional Analysis; Mathematical Methods.

Research Projects

2020 Co-Researcher, Academy of Scientific Research and Technology (ASRT), Egypt, Science UP Grant No. 6479.

Awards and Recognition

2019 Best Website Prize, Sohag University.

Technical Skills

Code Python (Intermediate/Advanced); Fortran (Intermediate).

Software LaTeX (Advanced); MATLAB (Advanced); Mathematica/Maple (Intermediate).

General Windows, Microsoft Office.

Tools

Languages

Arabic Native proficiency

English IELTS: 6.0

English Duolingo: 105

Selected Conferences and Academic Events

- (a) *MathConnect 2024: International Conference of Mathematics and its Applications, KFUPM, December 2024.*
- (b) *1st International Alumni's Mathematics UET Conference, UET Lahore (Pakistan), February 26, 2022.*
- (c) *Expert lecture on "National Mathematics Day," Anand International College of Engineering, Jaipur, India (Online), December 22, 2021.*
- (d) *Online International Workshop on "Modern Mathematical Methods and High-Performance Computing in Science & Engineering," India, November 23–24, 2021.*
- (e) *International Symposium on Recent Advances in Fractional Calculus, P. D. Patel Institute of Applied Sciences, CHARUSAT, Changa, Gujarat, India (Online), September 30–October 1, 2021.*
- (f) *International Conference on Applied Mathematics in Engineering (ICAME), Balikesir, Turkey (Online), September 1–3, 2021.*
- (g) *Symposium on Special Functions, Orthogonal Polynomials and Their Applications, Suez University, Egypt, 2021.*
- (h) *First Online Conference on Modern Fractional Calculus and Its Applications, Alexandria University, Egypt, 2020.*

Academic Service and Outreach

During the COVID-19 period, I maintained continuous academic communication with students and created a YouTube educational channel to support remote learning.

References

References available upon request.